

Unit 1. Introduction to DBMS

- **Data:** Anything can be called data.

Ex. Name of Person

Name of City

Number

- **Information:** When a data is meaningful is called Information.
- **Database:** It is an organized collection of related Information, because only organized collection of is database
Ex.: Dictionary
- **DBMS:** DBMS is nothing but Database Management System, It is particular piece of S/W that allow access to data content in Database.
- **Introduction to RDBMS:** A relational Database Management System is a program that lets you Create, Modify and administer a relational database.
Most Commercial RDBMS use the structured Query Language (SQL) to access data in the database.

1.1 Purpose/Advantages of DBMS: -

1. Reduction of data redundancy:

Duplication of data is called as redundancy, while creating a database it might be possibility record is repeated that repetition is called as redundancy.

In DBMS/RDBMS Overall data can be control by the DBA, so it avoids that unnecessary duplication of data.

2. Sharing of data:

DBMS allows multiple user to access the database at the same time. The control database is to store many users. This is concept of sharing of data.

3. Security:

Data is IMP part of any organization & may be confidential such confidential data must not be accessed by unauthorized persons or users.

The DBA is responsible for ensuring such kind of security has to design proper authentication method for access of RDBMS.

4. Integrity:

DBMS provides data integrity data integrity means data presentation database is both accurate & consistent data enter in database should be specified range of correct format.

Ex.: We can apply the constrains of database.

5. Centralize Control:

DBMS provide centralize control to database organize on may its data central & content database by database administrator.

The advantage of database, the centralize control is we can easily maintain backup & recovery.

1.2 Database System Applications: -

1. Universities: -

Universities have so much data which can be stored in the database, such as student information, teacher information, non-teaching staff information, course information, section information, and so on. University information is kept safe and secure in the database.

2. Banking: -

It is one of the major applications of databases. Banks have a huge amount of data as millions of people have accounts that need to be maintained properly. Banking databases store a lot of information about account holders. It stores customer details, banking transactions, balance sheets, credit card and debit card details, loans, fixed deposits, and so on. Everything is maintained with the help of a database.

3. Railway Reservation System: -

They store information such as passenger name, mobile number, booking status, reservation details, train schedule, employee information, account details, seating arrangement, route & alternate route details, etc. All the information needs to be maintained, so railways use a database management system for their efficient storage and retrieval purpose.

4. Social Media Sites: -

The various social media sites like Facebook, LinkedIn, Twitter, Instagram, etc. People can chat with their friends and family and make new friends from all over the world. Social media has millions of accounts, which means they have a huge amount of data that needs to be stored and maintained. Social media sites use databases to store information about users, images, videos, chats, etc.

5. Library Management System: -

There are hundreds and thousands of books in the library, so it is not easy to maintain the records of the books in a register or diary, so a database management system is used which maintains the information of the library efficiently. The library database stores information like book name, issue date, author name, book availability, book issuer name, book return details, etc.

Data Abstraction: -

Data Abstraction is a logical function in an application to separate the raw data from the front end. In simple terms, the data layer handles the connection to the database and querying it from the front end.

There are mainly 3 levels of data abstraction:

Physical or Internal Level

This is the lowest level of data abstraction. It tells us how the data is actually stored in memory. It will contains block of storages (bytes, GB,TB,etc)

Logical or Conceptual Level

This level comprises the information that is actually stored in the database in the form of tables. It will contain the fields and the attributes of data. At this level, the information available to the user at the view level is unknown.

View or External Level

This is the highest level of abstraction. Only a part of the actual database is viewed by the users. Users view data in the form of rows and columns. Tables and relations are used to store data. It works with GUI access of database.

Structure of DBMS: -

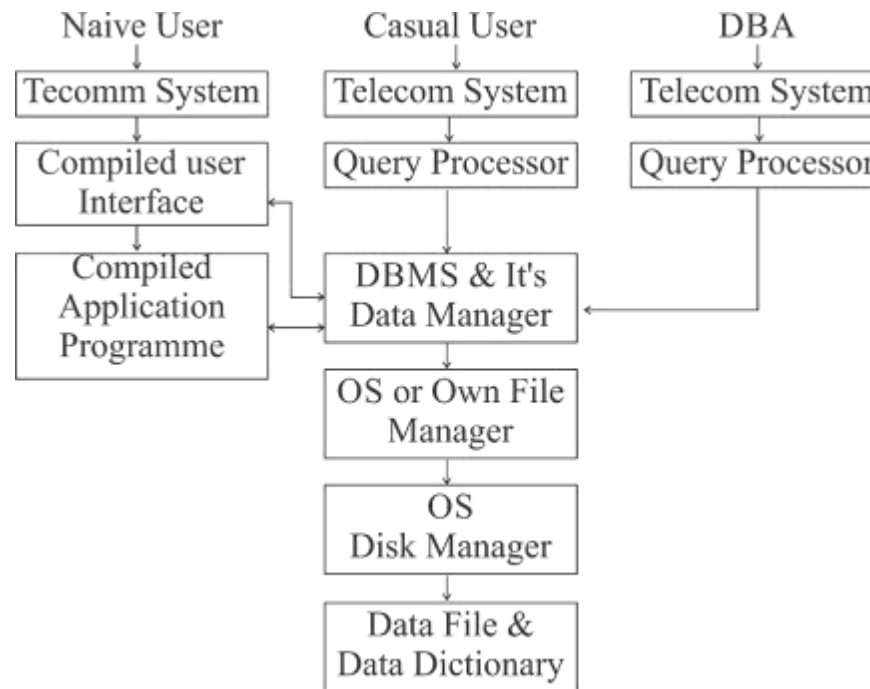


Fig. Structure of DBMS

DBMS is used to manage store & manipulate the data. Following are the users that wants in Database Management System.

1. **Naive User:** The Naive User interactive with the system by invoking application program or S/w have been previously written.
2. **Casual User:** The Casual User is the S/W Programmers or Professional Programmers. This is responsible for Application Programming & S/W.
3. **Database Manager:** It is simple called as Database Control System. It is also responsible for integrity & security, hacker, recovery of the data.
4. **File Manager:** The File Manager is responsible for structure of files. It is also responsible for transmitting the required record to the Data Manager.

5. **Disk Manager:** The Disk Manager is part of the operating System of the host Computer & all physical Input & Output operations are performed by it.
6. **Query Processor:** The Query processor is used to interact the online user's Query & Convert it into series of operations in a form capable of being sent to the Data Manager for evolution.
7. **Telecomm System:** Online users of a Computer System whether remote or local communicate with in by sending & receiving messages over communication lines. These messages are routed via an independent S/W System Called a Telecommunication System or a Telecommunication Control Program.
8. **Data Files:** Data files according to the database.
9. **Data Dictionary:** The Data Dictionary which is a database itself, documents the data.
10. **DBA (Database Administrator):** A person or the group of person, who controls the database system is called as DBA. It is responsible for creating Maintaining, Modifying the structure of Database System.

1.3 Relational Database Structure: -

1.Tables

The Foundation of a Relational Database at the core of a relational database are tables, also known as relations. Each table is a two-dimensional structure composed of rows and columns.

2.Primary Keys

Uniquely Identifying Rows In a relational database, each table should have a primary key, which is a unique identifier for each row to distinguish and Primary keys can be one or more columns.

3.Foreign Keys

Establishing Relationships Relational databases excel in managing relationships between tables. A foreign key is a column in one table that refers to the primary key in another table. It establishes a link between the two tables.

4.Normalization

Efficient Data Organization Normalization is a process used to organize data in a relational database efficiently. It minimizes data redundancy and ensures data integrity.

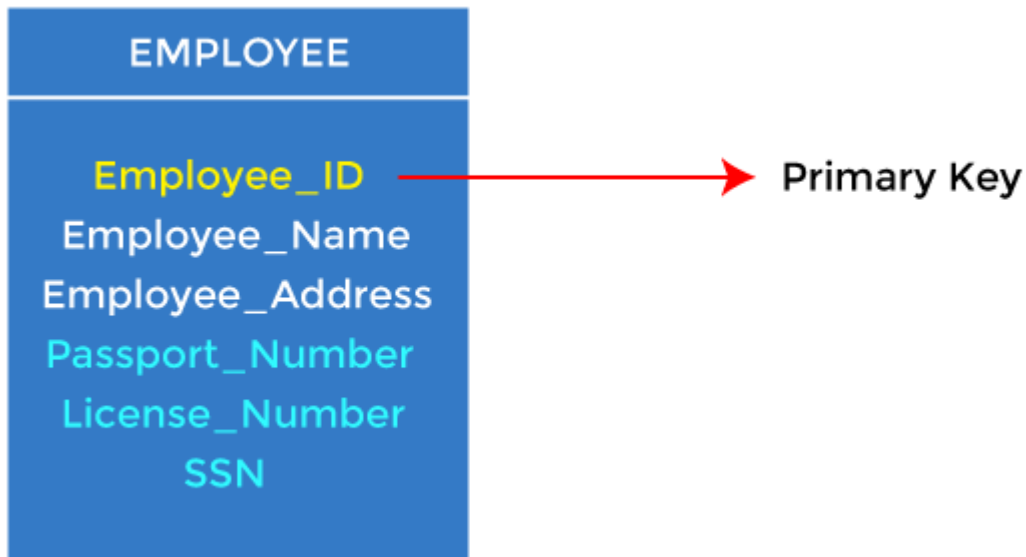
5.SQL

The Language for Database Operations Structured Query Language (SQL) is the standard language used to interact with relational databases. Common SQL commands include SELECT, INSERT, UPDATE, and DELETE.

1.4 Keys: -

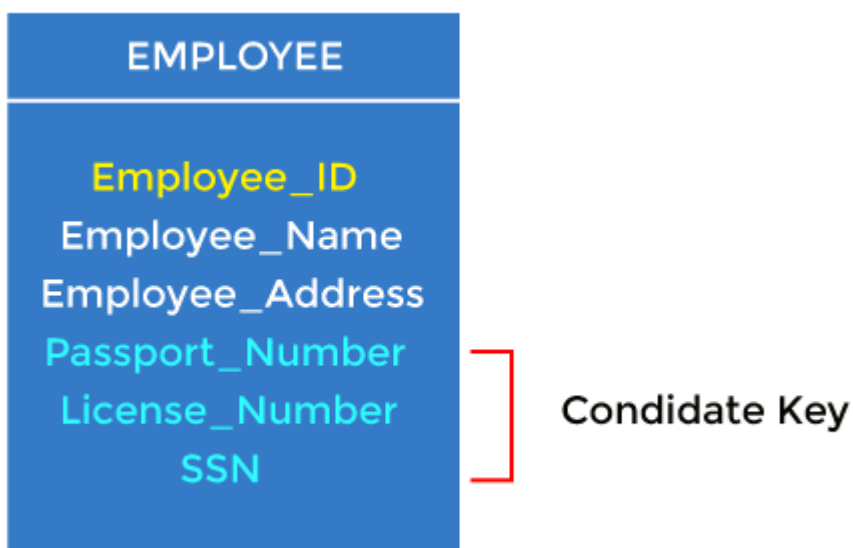
. Primary key

- It is the first key used to identify one and only one instance of an entity uniquely. The key which is most suitable from those lists becomes a primary key.



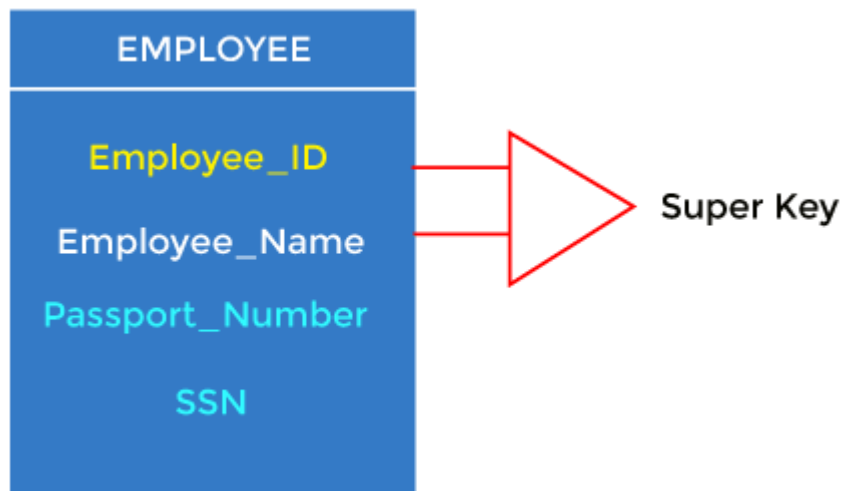
2. Candidate key

- A candidate key is an attribute or set of attributes that can uniquely identify a tuple.
- Except for the primary key, the remaining attributes are considered a candidate key. The candidate keys are as strong as the primary key.



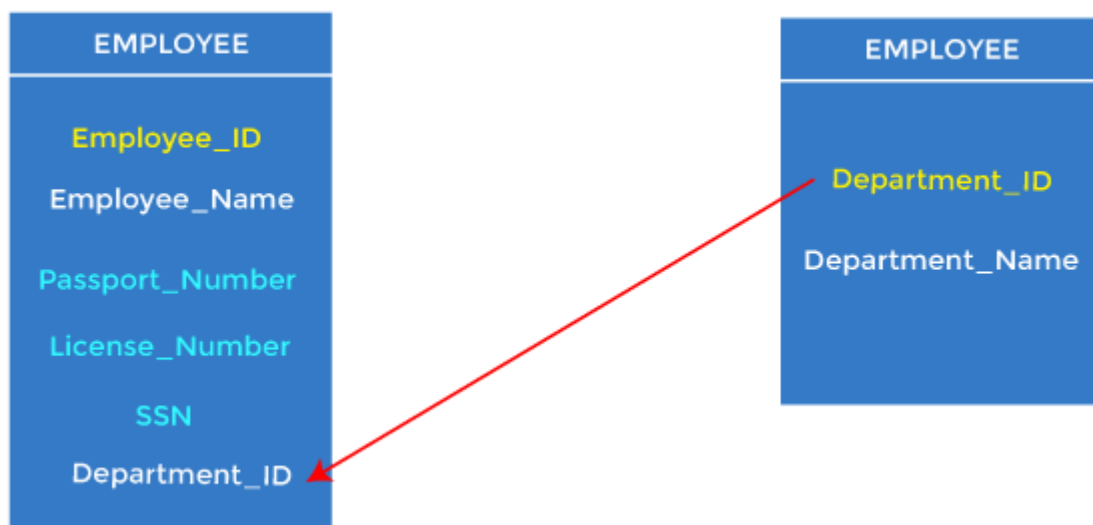
3. Super Key

Super key is an attribute set that can uniquely identify a tuple. A super key is a superset of a candidate key.



4. Foreign key

- Foreign keys are the column of the table used to point to the primary key of another table.



1.5 Relational Algebra: -

Relational algebra is a collection of operations that used to manipulate the relations.

The relational algebraic operations are divided into two parts:

1. Basic set oriented operations, that contains

a. Union:

In this operation represent set or relation contains the tuples that are either in the first relation or in second relation at in both of them. Duplications eliminated in this operation.

Ex. Consider following two tables:

P	
Id	Name
101	Vaibhav
102	Akshay
103	Arun
104	Sudhir

Q	
Id	Name
101	Vaibhav
105	Vaijinath
106	Tanaji
102	Akshay

If P(P) & Q(Q) are the two compatible relations. The Union of P & Q can be calculated as follow:

$$R = P \cup Q$$

Or

$$R = \{t / t \in p \vee t \in Q\}$$

Following table or relation shows the O/P of $P \cup Q$

R	
Id	Name
101	Vaibhav
102	Akshay
103	Arun
104	Sudhir
105	Vaijnath
106	Tanaji

b. Set the difference operation:

The difference operation removes common on tuples from first relation.

It allows find tuples that are available in first relation but not in other relation.

Ex.:

P	
Id	Name
101	Vaibhav
102	Akshay
103	Arun
104	Sudhir

Q	
Id	Name
101	Vaibhav
105	Vaijinath
106	Tanaji
102	Akshay

This relation represents as:

$R = P - Q$

Or

$R = \{t / t \in p \wedge t \notin Q\}$	
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The O/P is

R	
Id	Name
103	Arun
104	Sudhir

c. Intersection operation:

The intersection operation success a display the common tuples from the two relation.

This operation is represented as Ex.:

P	
Id	Name
101	Vaibhav
102	Akshay
103	Arun
104	Sudhir

Q	
Id	Name
101	Vaibhav
105	Vaijinath
106	Tanaji
102	Akshay

$$R = P \cap Q$$

Or

$$R = \{t / t \in p \wedge t \in Q\}$$

The O/P is,

R	
Id	Name
101	Vaibhav
102	Akshay

b. Cartesian product:

The Cartesian product of two relations is the concatenation of tuples belonging to the two relation a new resultant relation's created.

Which is having all possible concatenation of tuples.

Ex: -

Emp		Sal
Id	Name	Salary
e1	Amit	5000
e2	Amol	6000
e3	suman	7000

The cartesian product can be represented as ,

$$R = \text{Emp} \times \text{Sal}$$

or

$$R = \{t1 \parallel t2 \mid t1 \in \text{Emp} \wedge t2 \in \text{Sal}\}$$

Here the result is obtained by containing each tuple in all relation Emp with each tuple in relation salary.

Relational Database: -

1. Entity:

An entity can be a real-world object that can be easily identifiable.

For Ex.: In a school database, students, teachers, classes & courses offered can be considered as entities.

All these entities have some attributes or properties that give them their identity.

2. Attribute:

Each object or entity having some property called as attribute.

In the file system, the file is used to specify properties, the object in RDBMS attributes corresponded to the file.

Ex: The object person has properties like age, color & so on.

3. Domain:

Set of values of same data type is called as Domain.

The domain is set of homogenous members. A domain can be defined as D.

Tuples:

Object entity having attributes can be presented by an ordered set called as tuples.

Assume that n attributes take values from the domain D1, D2.Dn.

An resulting set of Cartesian product of domain contains all possible N tuples. Ex.

Applicant

Name	Age	Profession
Ajit	20	S/W developer.

The applicant object can be represent by following values

[Ajit, 20, S/W develop]

This possible order obtained from Cartesian product of domain for attributes, name, age & profession is called as tuple.

1.6 Basic Concept of ER Model: -

Entity-relationship Model

- An Entity is a thing / object which is different from other thing / object. OR Entities are distinguishable objects. An entity is characterized by its properties called attributes. For Ex Student can have Roll_No, Name, Class...Percent etc
- In this database model, relationships are created by dividing object of interest into entity and its characteristics into attributes. Different entities are associated using relationships.
- For example, in the following fig we have two entities Student, College & their relationship. The relationship between Student & College is Many-to-many as a college can have many students however, a student cannot study in multiple colleges at the same time. Student entity has attributes such as SID, SNAME & SADD and College entity has attributes such as COL_ID, COL_NAME.

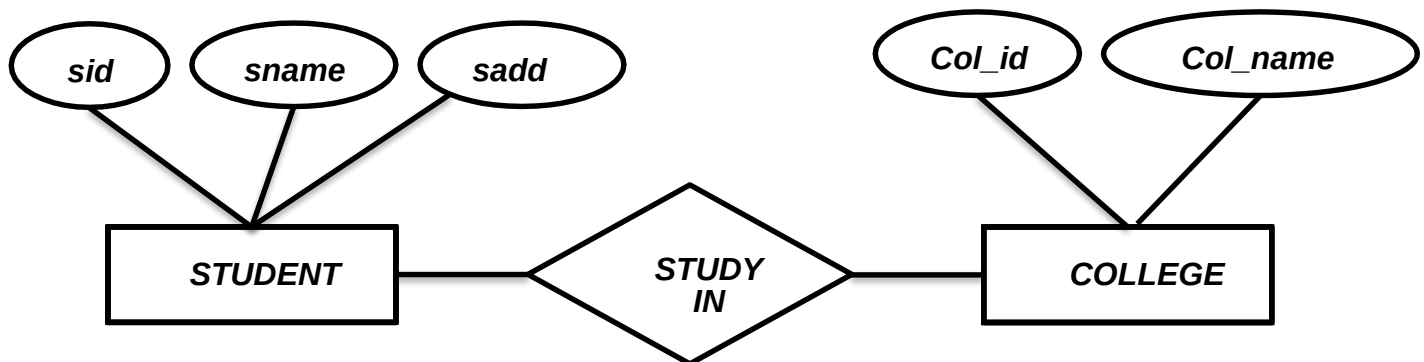


Fig E-R model

- Here are the geometric shapes & their meaning

Shape	Meaning in ER Data Model
Rectangle	Represent Entity/entity set
Ellipse	Attributes
Diamond	Relationship/Relationship Set
Lines	To link attributes to entity & entity to relationship

1.7 Entity Sets, Relationship Sets, Weak Entity Sets: -

Entity Set is a collection of entities of the same entity type.

1. Strong Entity Set: -

A strong entity set is an entity set that contains sufficient attributes to uniquely identify all its entities.

In other words, a primary key exists for a strong entity set.

Primary key of a strong entity set is represented by underlining it.

2. Weak Entity Set: -

A weak entity set is an entity set that does not contain sufficient attributes to uniquely identify its entities.

In other words, a primary key does not exist for a weak entity set.

However, it contains a partial key called as a discriminator.

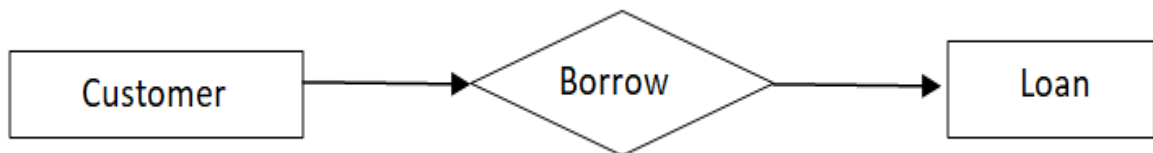
Discriminator can identify a group of entities from the entity set.

What is a Relationship Sets?

It is a set of Relationships of the Same type. Mathematical Relation on $m > 2$ (Possibly non-distinct) entity Sets.

If $E_1, E_2, \dots, E_n$ are the entity sets for the Relationship Set R is a subset of $\{(e_1, e_2, \dots, e_n) \mid e_1 \in E_1, e_2 \in E_2, \dots, e_n \in E_n\}$ Where $(e_1, e_2, \dots, e_n)$ is a Relationship.

For Example, (E-R Diagram)



Relationship Set

Characteristics of Relationship Set

Degree: Degree of a relationship set indicates to the number of properties related with the relationship set.

Arity: Arity of a relationship set indicates the number of taking part relations. It can be like double (including two relations), ternary (including three relations), and so on.

Cardinality: Cardinality characterizes the number of occurrences or records that can be related with each substance on both sides of the relationship.

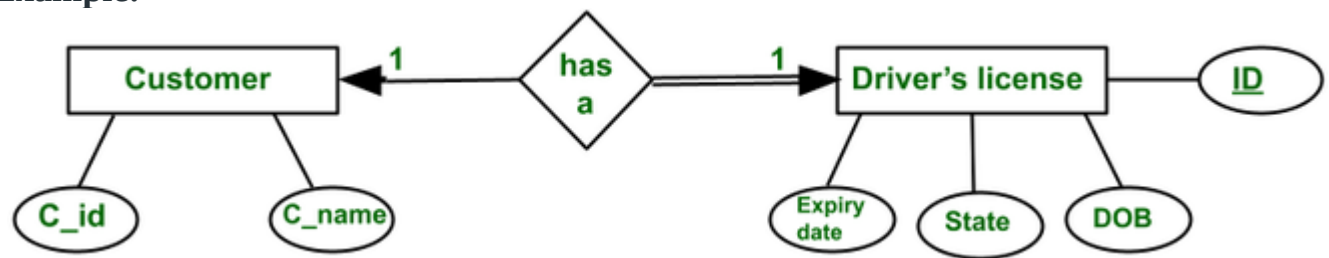
1.8 Mapping Cardinalities , E-R Diagrams: -

In a **Database Management System (DBMS)**, mapping cardinalities explains how entities (tables) in a database schema relate to one another.

1. One to one relationship (1:1)

It is represented using an arrow($\rightarrow$, $\leftarrow$)(There can be many notations possible for the ER diagram).

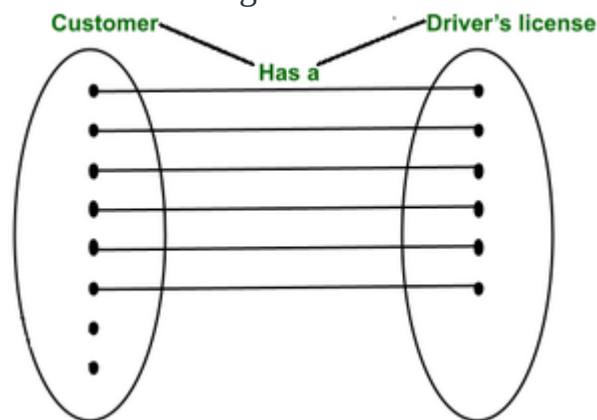
Example:



One to One relationship

In this ER diagram, both entities customer and driving license having an arrow which means the entity Customer is participating in the relation “has a” in a one-to-one fashion. It could be read as ‘Each customer has exactly one driving license and every driving license is associated with exactly one customer.

The set-theoretic perspective of the ER diagram is

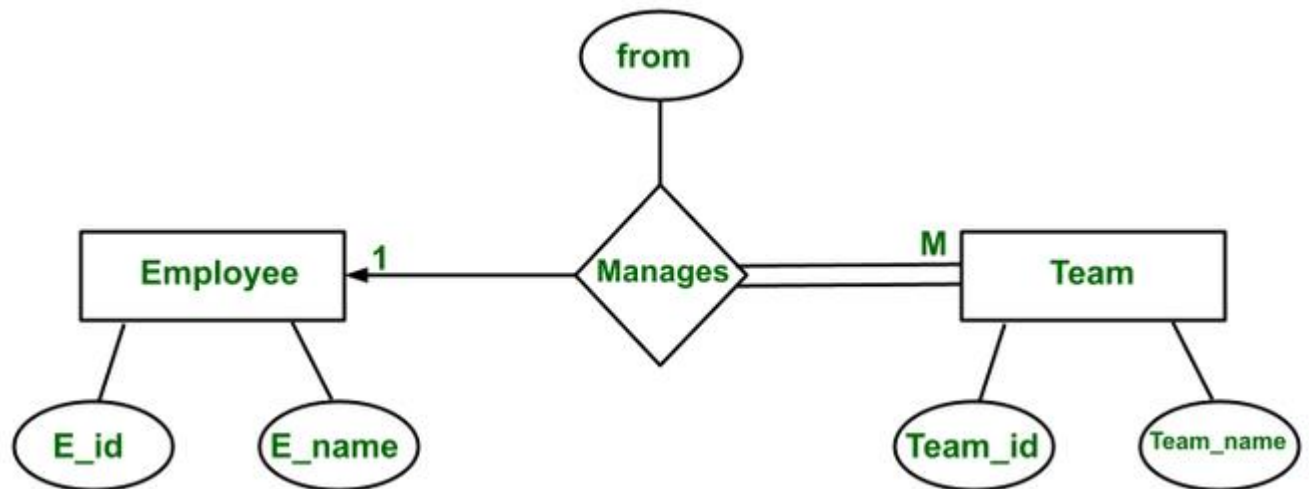


ER diagram

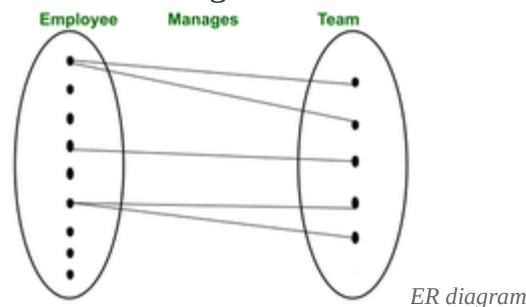
There may be customers who do not have a credit card, but every credit card is associated with exactly one customer. Therefore, the entity customer has total participation in a relation.

2. One to many relationship (1:M)

Example:

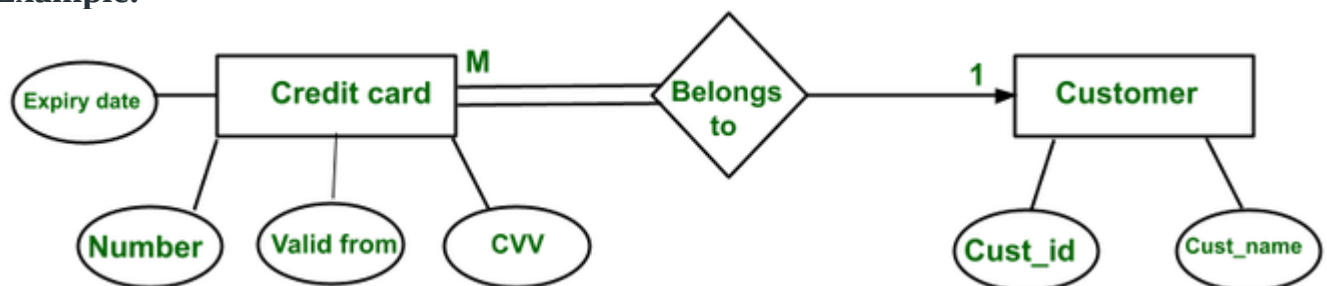


This relationship is one to many because “There are some employees who manage more than one team while there is only one manager to manage a team”. The set-theoretic perspective of the ER diagram is:



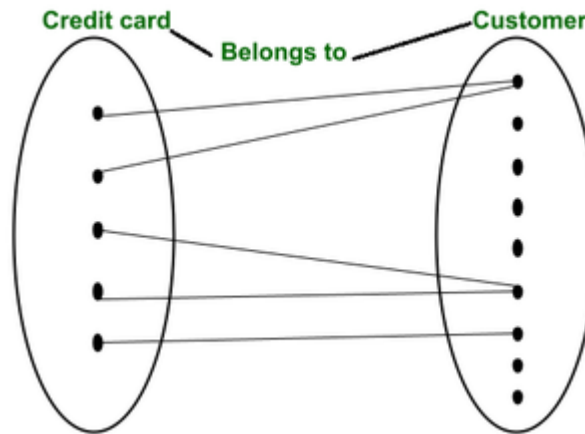
3. Many to one relationship (M:1)

Example:



It is related to a one-to-many relationship but the difference is due to perspective. Any number of credit cards can belong to a customer and there might be some customers who do not have any credit card, but every credit card in a system has to be associated with an employee(i.e. total participation). While a single credit card can not belong to multiple customers.

The set-theoretic perspective of the ER diagram is:

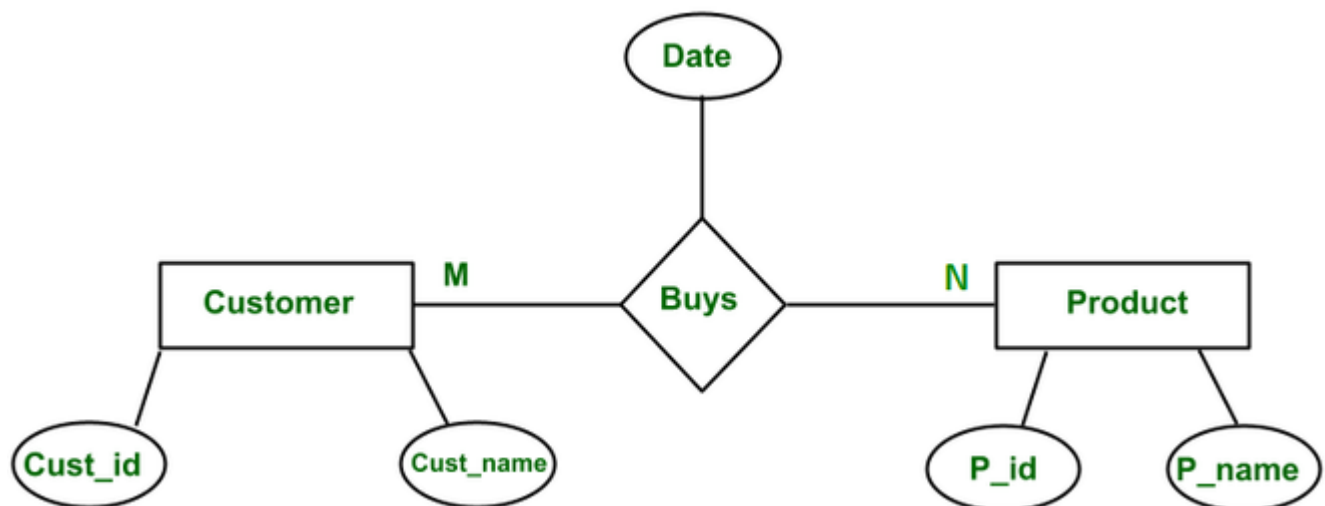


ER diagram

4. Many to many relationship (M:N)

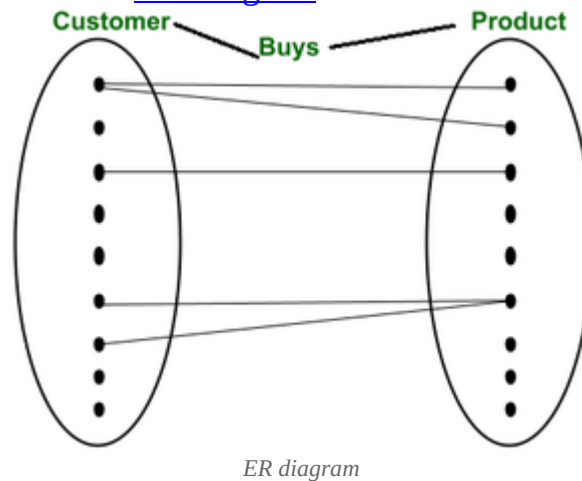
Example:

A customer can buy any number of products and a product can be bought by many customers.



Many to many Relationship

The set-theoretic perspective of the [ER diagram](#) is:



Any of the four cardinalities of a binary relationship can have both sides partial, both total, and one partial, and one total participation, depending on the constraints specified by user requirements.

Extended E-R Features: -

Subtypes and Supertypes:-

Extended ER models introduce the concept of subtypes and supertypes, allowing for the representation of hierarchical relationships within an entity. This is particularly useful when dealing with entities that have multiple levels of specialization. For example, in a university database, “Person” may be a supertype, and “Student” and “Faculty” could be subtypes.

Inheritance: -

Inheritance is closely related to subtypes and supertypes. It allows attributes and relationships defined in a supertype to be inherited by its subtypes. In our university example, common attributes like “Name” and “Address” can be inherited by both students and faculty.

Union Types: -

Extended ER models introduce union types, which allow an entity to have multiple types simultaneously. For instance, an employee in an organization can have roles as both a “Manager” and a “Developer.”

Aggregation: -

It allows entities to be composed of other entities, forming a whole-part relationship. For instance, a “Department” entity can be composed of “Employee” entities, representing the employees working in that department.

Derived Attributes: -

Derived attributes are computed from other attributes in the database. They do not need to be stored explicitly but can be calculated when needed. For example, in a “Person” entity, the “Age” attribute can be derived from the “Date of Birth” attribute.

Constraints: -

Extended ER models allow for the specification of various constraints, including uniqueness constraints, inclusion constraints, and exclusion constraints. These constraints ensure data integrity and accuracy within the database.

